

A HIGH ENERGY DESIGN FOR JLEIC ION COMPLEX

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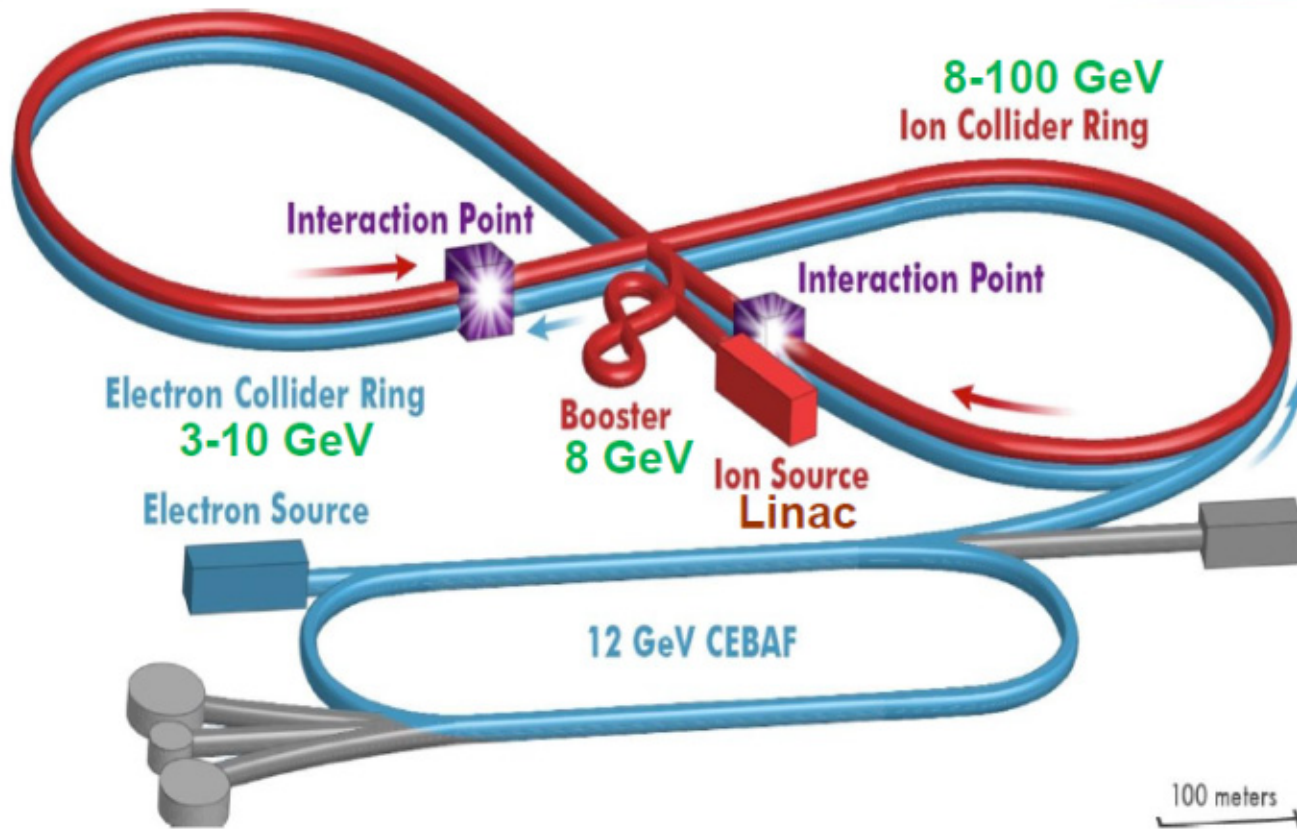
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P. Ostroumov and A. Plastun (now FRIB)**

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Outline

- ❑ Background / Motivations for Alternative Design Studies
- ❑ Review: Alternative Design for the 65-GeV CM Option
- ❑ The High-Energy Design - 100 GeV CM or higher
- ❑ Racetrack Pre-Booster Ring
- ❑ Higher Energy Ion Booster, could serve as e-Ring
- ❑ Conclusions

Starting Point: JLEIC 65-GeV Baseline Design



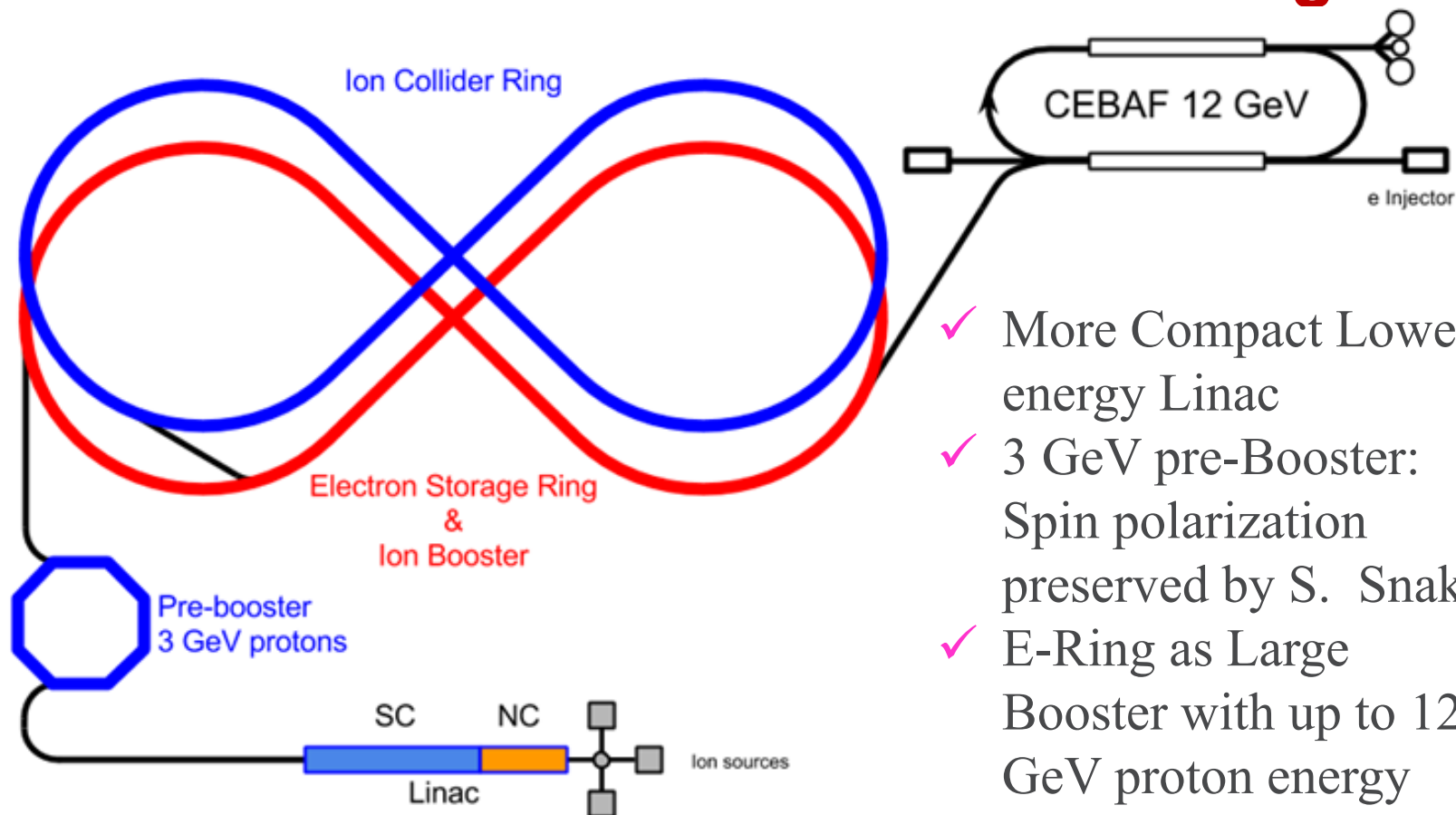
- ✓ Pulsed SC Linac up to 280 MeV p - 100 MeV/u Pb
- ✓ 8 GeV Booster followed by 100 GeV ion Collider Ring
- ✓ CEBAF is full energy injector for the e Collider Ring
- ✓ All rings are figure-8 to preserve spin polarization

Main Motivations for the Alternative Design Approach

- ❖ The idea is not to replace the baseline design but rather to investigate alternative options for the different components of the ion complex that have the potential of lowering the cost, mitigating the risk, and be ready for possible staging or future upgrades of the project
- ❑ Reduce footprint and cost of the Ion Complex
 - Compact Injector Linac (140 MeV instead of 280 MeV protons)
 - Small Pre-booster Ring (3-5 GeV, no figure-8)
 - Consolidate Electron Ring as Large Ion Booster
- ❑ Lower the risk
 - Use RT magnets whenever possible
 - Avoid transition crossing for all ions
- ❑ Staging / Upgrade
 - Ion Injector Complex Compatible with 65, 100 and 140 GeV CM
 - Only Upgrade Ion Collider Ring

Review of the 65-GeV Design

Review: Alternative for the 65-GeV Design

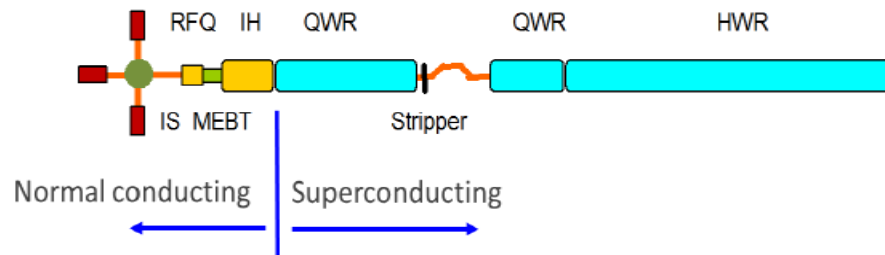


- ✓ More Compact Lower-energy Linac
- ✓ 3 GeV pre-Booster: Spin polarization preserved by S. Snakes
- ✓ E-Ring as Large Booster with up to 12 GeV proton energy
- ✓ Separate RF sections for electrons and ions ...

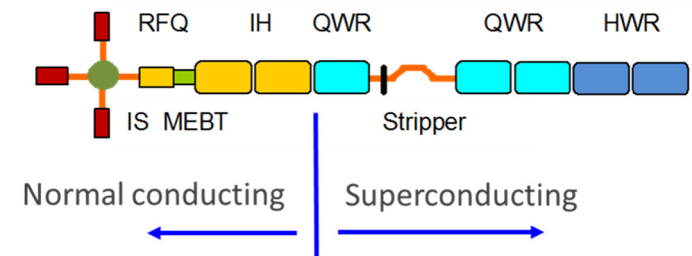
- ✓ The E-Ring and Ion Collider Ring are stacked vertically in one tunnel
- ✓ Ion injection from the E-Ring to Ion Ring is a vertical bend

A More Compact Lower-Energy Linac

Original



New



Item / Parameter	Original	New	Comments
Protons (MeV)	280	135	
Pb (MeV/u)	100	42	
SC modules	16	5	~ 1/3 Cost
Total Length	130	55	~ 1/2 Tunnel cost
Total Power (kW)	560	260	~ 1/2

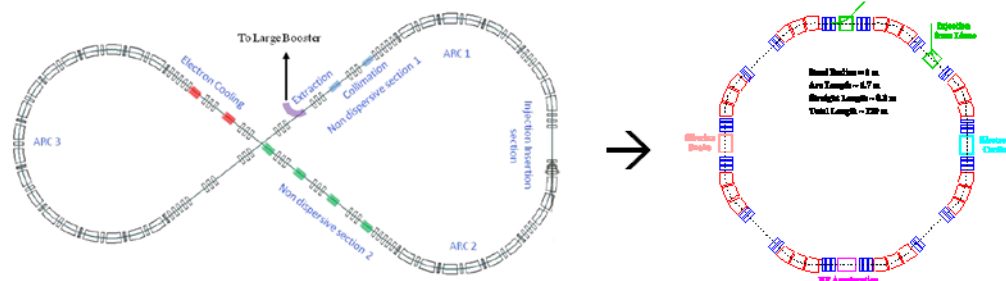
- Shorter Linac with higher performance cavities in pulsed mode
- The new design ~ 1/3 the construction cost of the original

Ref: P. Ostroumov et al, HB-2016, Malmo, Sweden

A More Compact Pre-Booster Ring

Original 3-GeV Pre-Booster

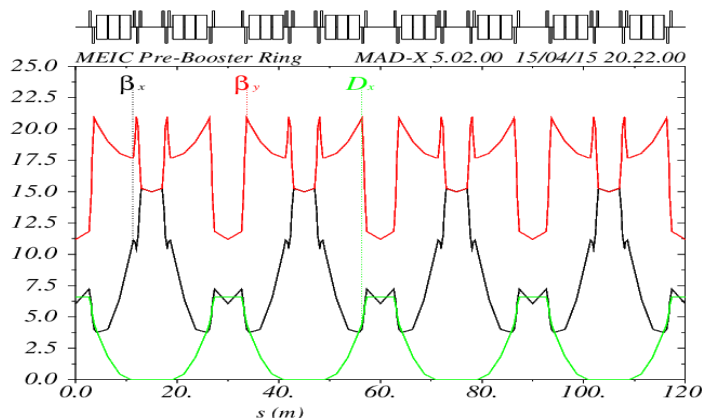
New Design



Item / Parameter	Original	New
N. of 15° Dipoles	36	24
N. of Quads	95	40
Total N. of Magnets	131	64
Total Length	234	120

➤ The new design ~ 1/2 the cost of original

Beam Optics



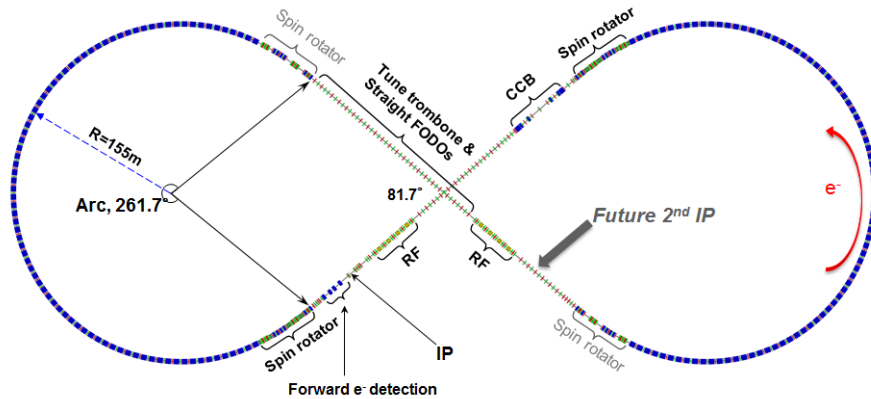
Ref: B. Mustapha et al, NAPAC-16, Chicago, IL

Design Parameters

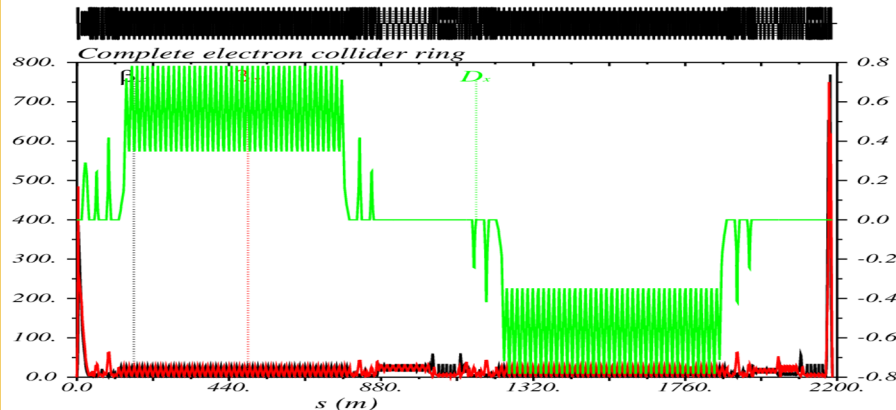
Parameter	Octagonal
Circumference, m	120
Arc length, m	6.7
Straight section length, m	8.3
Maximum β_x	15.3
Maximum β_y	21.0
Maximum dispersion	4.2
β_x at injection	6.0
Normalized dispersion at injection: $D/v \beta_x$	1.71
Tune in X	3.01
Tune in Y	1.18
Gamma transition	4.7
Gamma at extraction (3 GeV)	4.22
Momentum compaction factor	0.045
Number of quadrupoles	40
Quadrupole length, m	0.4
Quadrupole half aperture, cm	5
Maximum quadrupole field, T	1.5
Number of dipoles	24
Dipole bend radius, m	8
Dipole angle, deg	15
Dipole full gap, cm	5
Maximum dipole field	1.6

E-Ring As Large Booster for the Ions – Added RF Sections for Ions Acceleration ...

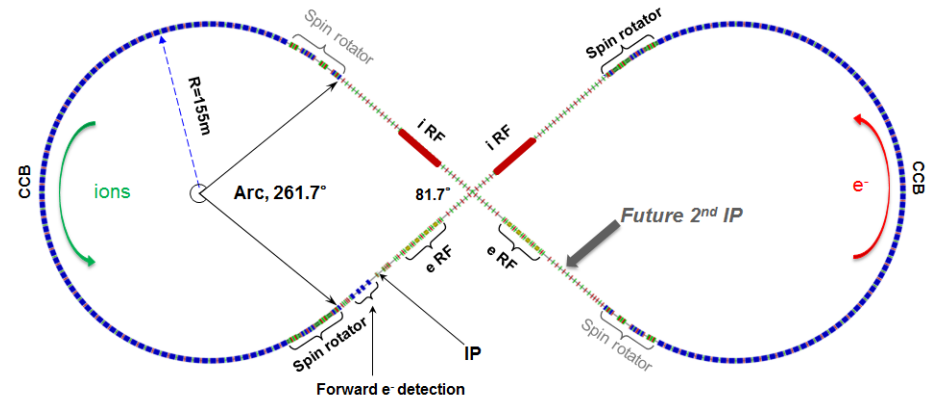
Electrons Only



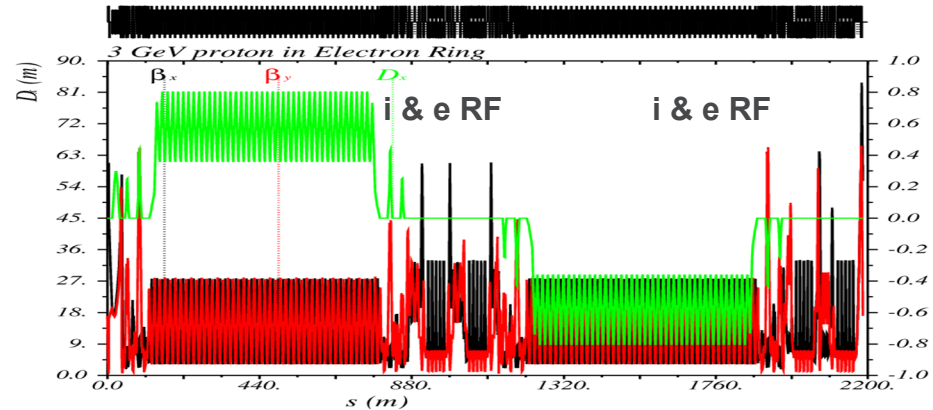
e- Collision Optics at 5 GeV



Electrons & Protons



Proton optics at 3 GeV, Injection from Booster

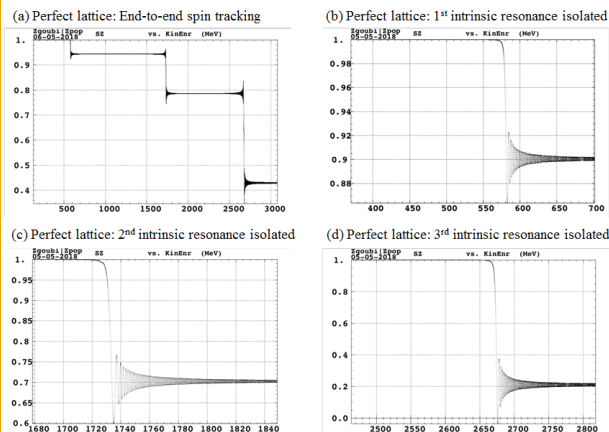


- ✓ Ion RF sections were inserted in the straight sections, across from electron RF
- ✓ Proton beam optics studied at the injection energy of 3 GeV

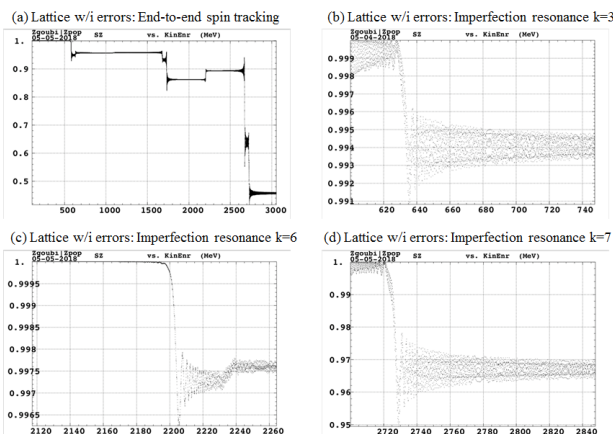
Ref: B. Mustapha et al, IPAC-17, Copenhagen, Denmark

Spin Dynamics in 3-GeV Pre-Booster

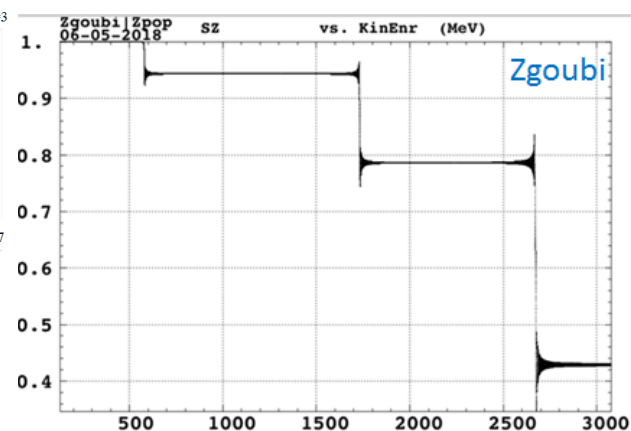
Intrinsic resonances (proton)



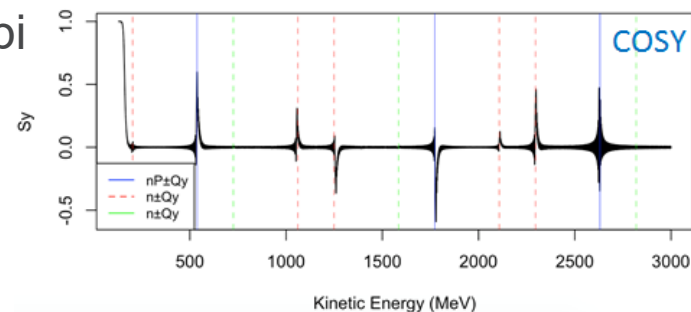
Imperfection resonances (proton)



Intrinsic : COSY vs. Zgoubi



- ✓ Good overall agreement between COSY and Zgoubi
- ✓ No resonances observed for deuterons, first one expected at ~ 5.6 GeV/u
- ✓ Possible spin correction schemes for protons are listed in table below



Option	~ 5 Imperfection	~ 2 Strong Intrinsic	~ 1 Intrinsic	~ 8 Weak Intrinsic
A	Orbit corrections	Rf Dipole	Rf Dipole	Nothing/Pulsed Quads
B	5% Siberian Snake	Rf Dipole	Rf Dipole	Nothing/Pulsed Quads
C	Orbit Correction	Pulsed Quads	Pulsed Quads	Nothing/Pulsed Quads
D	5% Siberian Snake	Pulsed Quads	Pulsed Quads	Nothing/Pulsed Quads
E	40% Siberian Snake	40% Siberian Snake	40% Siberian Snake	40% Siberian Snake

Ref: J. Martinez et al, IPAC-18, Vancouver, Canada

Beam Formation in Alternative 65-GeV Design

Proton beam parameters in the Rings

Parameter/Ring	Pre-Booster	Booster: E-Ring	Collider Ring
Inj. energy [GeV]	0.13	3	16
End energy [GeV]	3	16	100
Circumference [m]	120	2250	2250
Harmonic	1	18	18
Injection efficiency	90	100	100
Bunch charge [μC]	0.21	0.21	0.21
Ions per bunch [10^{12}]	1.32	1.32	1.32
RMS norm. emittance [$\pi\text{-mm-mrad}$]	3	3	1
Bunch length [m]	84	84	84
SC tune shift at inj.	0.122	0.083	0.013
RF freq. at inj. [MHz]	1.21	2.322	2.394
RF freq. at end [MHz]	2.43	2.394	2.398
RF period at end [μs]	0.41	0.42	0.42
Avg. current at end [A]	0.51	0.5	0.5

Lead beam parameters in the rings

Parameter/Ring	Pre-Booster	Booster: E-Ring	Collider Ring
Inj. energy [GeV/u]	0.04	0.61	16
End energy [GeV/u]	0.61	16	40
Circumference [m]	120	2250	2250
Harmonic	1	18	18
Charge state [e]	67	82	82
Injection efficiency [%]	70	75	100
Bunch charge [μC]	0.23	0.21	0.21
Ions per bunch [1010]	2.1	1.6	1.6
RMS norm. emittance [$\pi\text{-mm-mrad}$]	3	3	1
Bunch length at inj. [m]	84	84	84
SC tune shift at inj.	0.086	0.256	0.005
RF freq. at inj. [MHz]	0.74	1.844	2.394
RF freq. at end [MHz]	2.03	2.394	2.398
RF period at end [μs]	0.49	0.42	0.42
Avg. current at end [A]	0.47	0.5	0.5

- ✓ The target luminosity of $10^{34} \text{ cm}^{-2} \text{ s}^{-1}$ was maintained
- ✓ The main issue is the high SC tune shift for Pb ions in the E-ring / Large Booster
- ✓ A 5-GeV or higher booster energy will reduce it to below 0.15

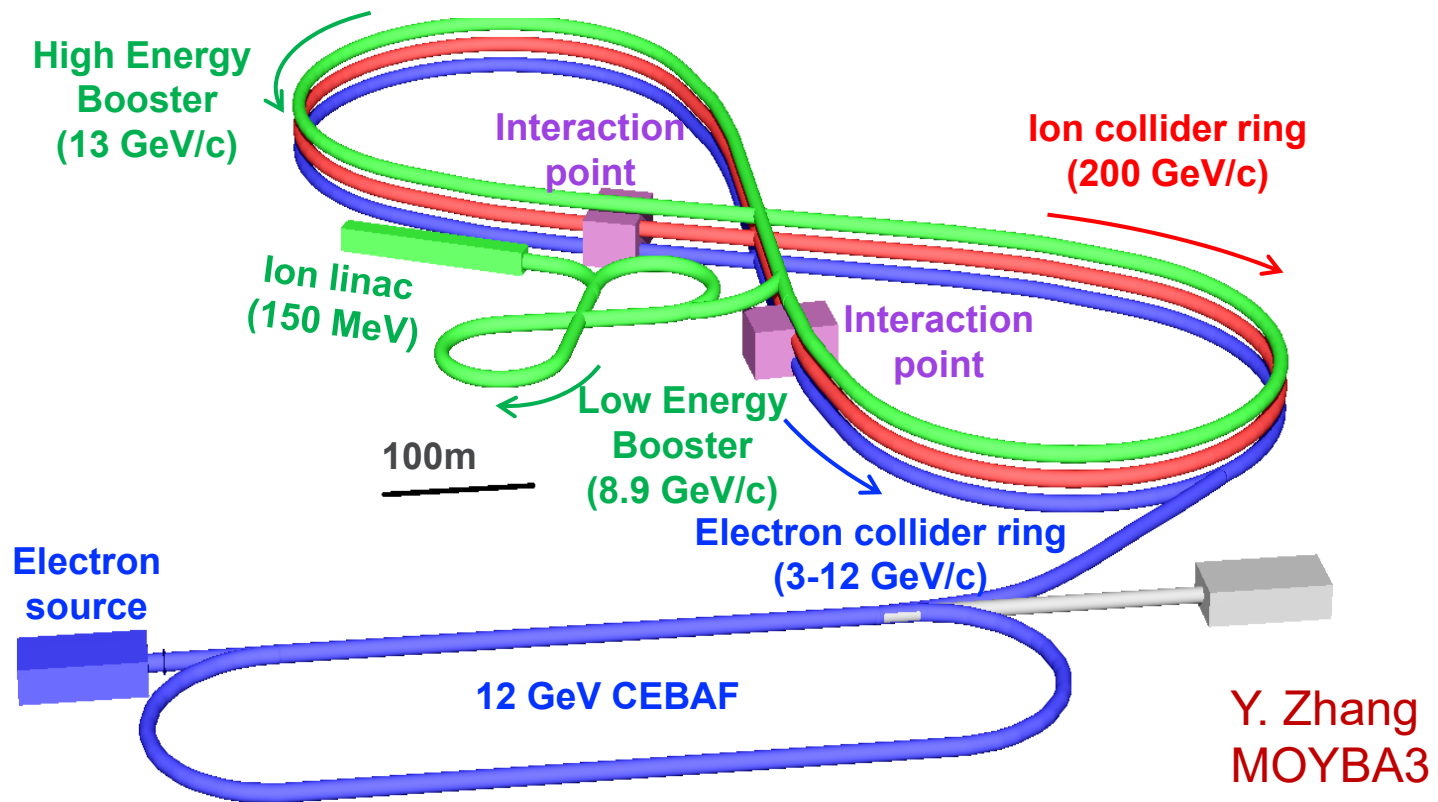
Ref: B. Mustapha et al, IPAC-18, Vancouver, Canada

Main Conclusions for the 65-GeV Design ...

- ☐ Lower linac energy seems reasonable for both proton and heavy-ion injection into a small pre-booster ring
- ☐ Higher energy pre-booster of 5 GeV or higher is required to lower SC tune shift for heavy ions in the e-ring
- ☐ E-ring as Large Ion Booster seems feasible from space and beam optics point of view

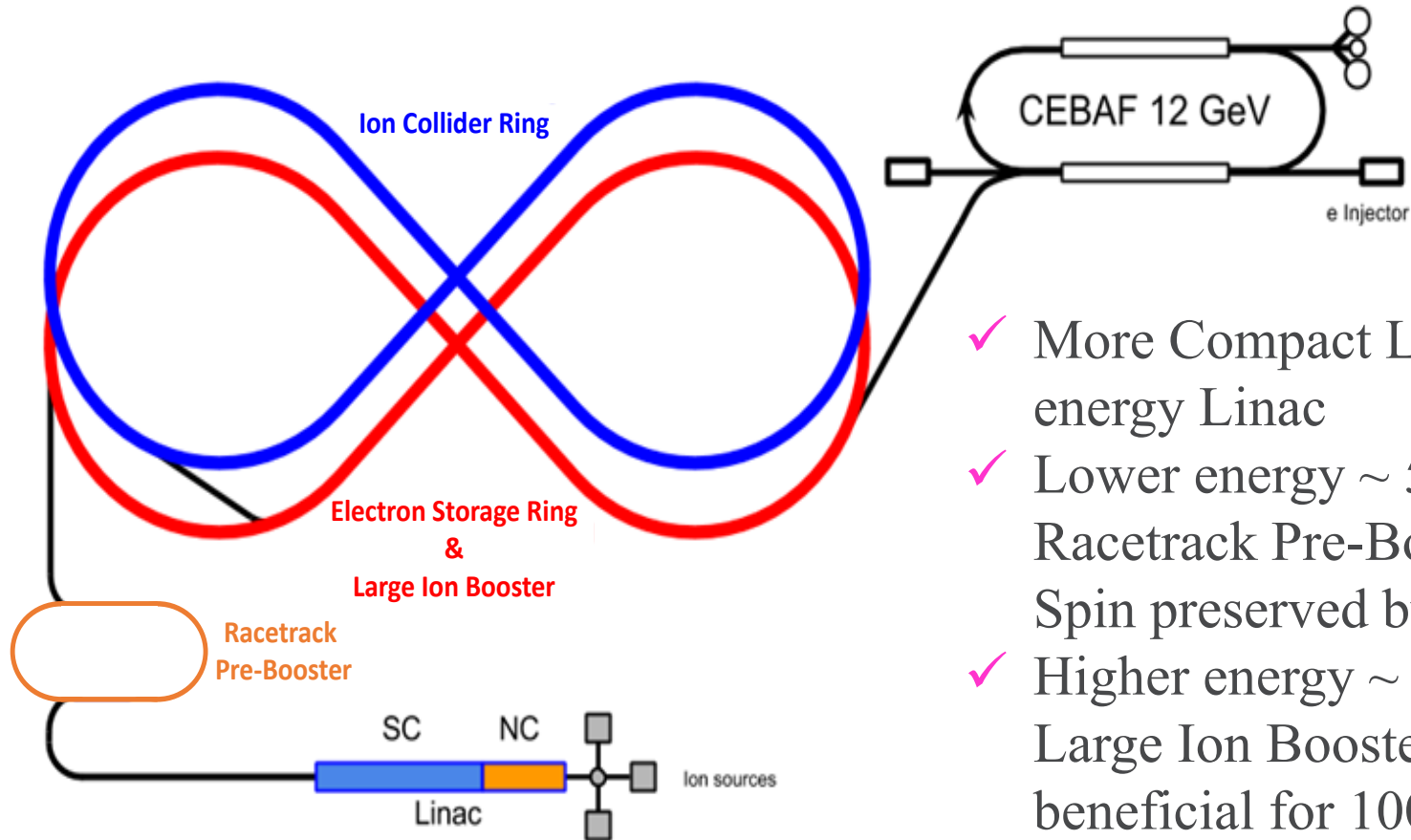
The New 100-GeV Design

The JLEIC 100-GeV Baseline Design



- ❑ Changes adopted from the alternative design approach
 - Lower-energy, shorter linac → 150 MeV
 - Two boosters: 8 and 13 GeV
 - RT magnets in the boosters, SC only in collider ring

Alternative Approach for 100-GeV or higher

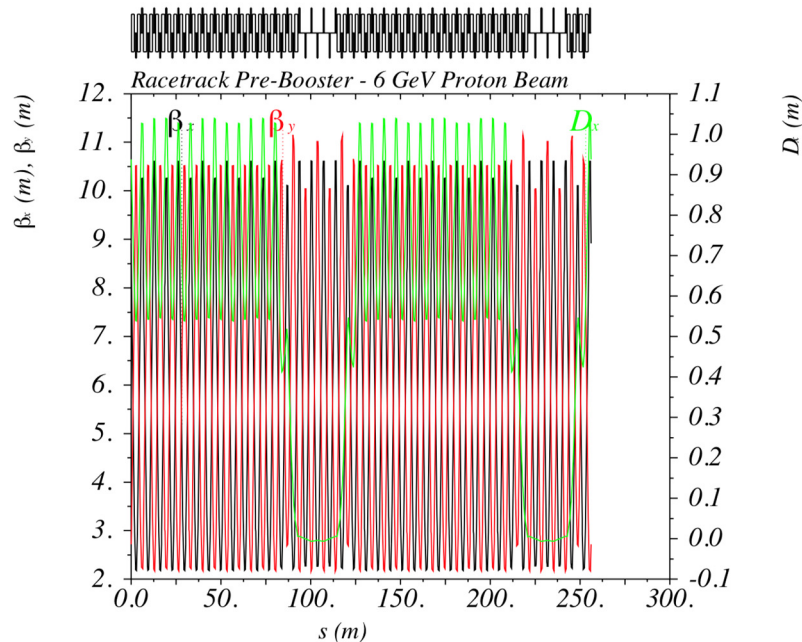


- ✓ More Compact Lower-energy Linac
- ✓ Lower energy ~ 5 GeV Racetrack Pre-Booster: Spin preserved by Snakes
- ✓ Higher energy ~ 16 GeV/u Large Ion Booster is beneficial for 100 GeV collider & future upgrade
- ✓ Large Ion Booster can serve as e-Ring ...

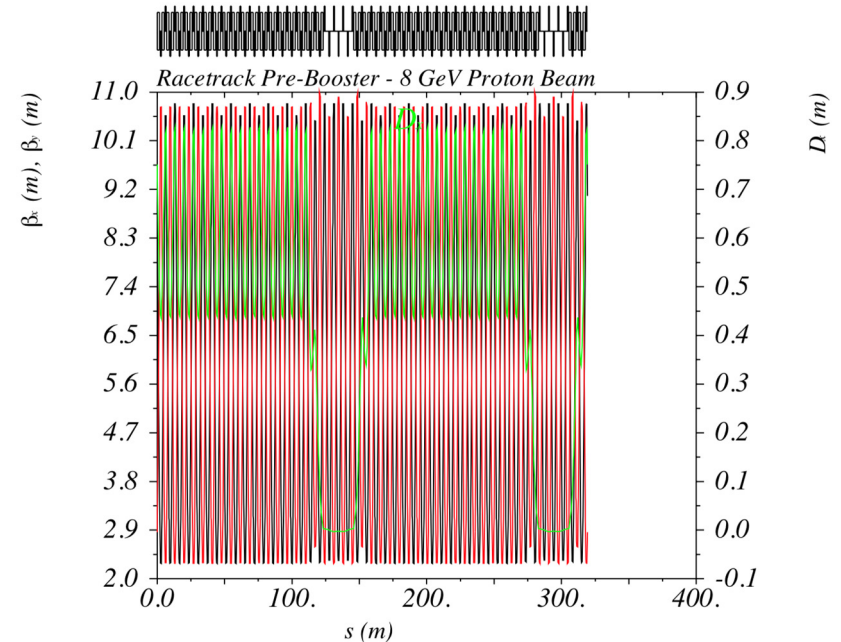
- ✓ Baseline e-Ring lattice can't be used as high energy ion Booster
- ✓ Need a new Booster/e-Ring design ...

Pre-Booster Design Options – 6 and 8 GeV

6 GeV proton beam optics



8 GeV proton beam optics



□ Design Considerations & Choices

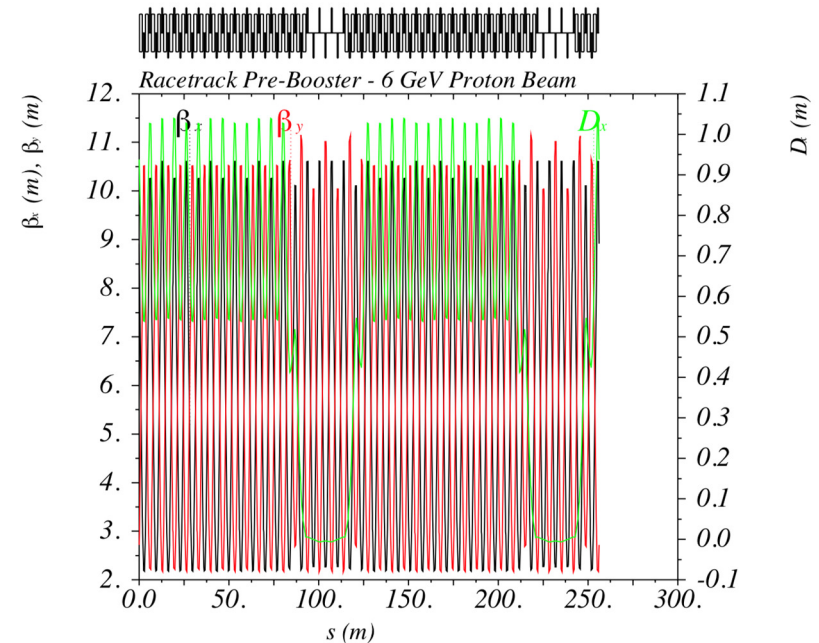
- Racetrack instead of figure-8, reduced circumference, spin OK
- Beam size and aperture in the large booster (injection energy)
- SC tune shift for protons and heavy ions
- Beam formation (cycles) and cooling requirements

Pre-Booster Design Choice: 6-GeV

All RT Race-Track Design

Parameter	Value
Circumference, m	256.1
Maximum β_x , m	10.6
Maximum β_y , m	11.14
Maximum dispersion, m	1.04
Tune in X	9.45
Tune in Y	9.36
Extraction γ	7.39
Transition γ	7.91
Quad. half aperture, cm	5
Quad. max. grad. T/m	20
Quad. length in arc, m	0.56
Dipole max. field, T	1.6
Dipole Length, m	1.51

6 GeV proton beam optics



Siberian snake for p, 3He
No resonances for d

Design of a High-Energy Large Booster / E-Ring

□ Design Goals & Constraints

- Avoid transition crossing for all ions ~ 16 GeV/u
- All RT magnets: 1.6 T max for dipoles, 20 T/m max for quads
- Shape & Size: figure-8, cross angle similar to collider ring
- Can be used as electron storage ring

□ Strong focusing required for high γ_{tr} $\rightarrow$ Investigated Options

- Combined function magnets
- Long quadrupoles in FD lattice (twice as long as e-Ring quads)
- Quadrupole doublets in FFDD lattice (may use PEP-II quads)

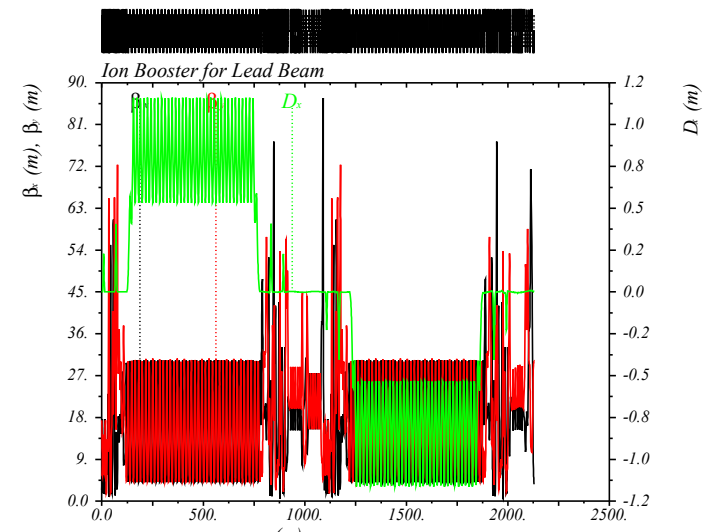
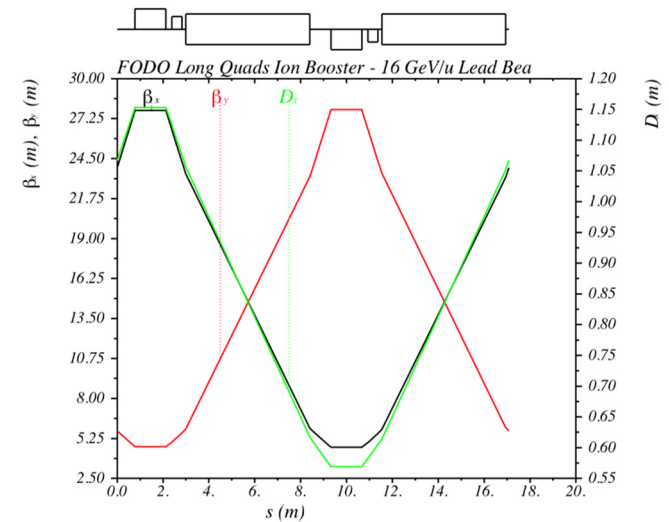
□ See TUPLO01 by J. Martinez, “Dual-Function Electron Ring / Ion Booster Design for JLEIC High-Energy Option”

Large Booster Design Choice: Long Quads Option

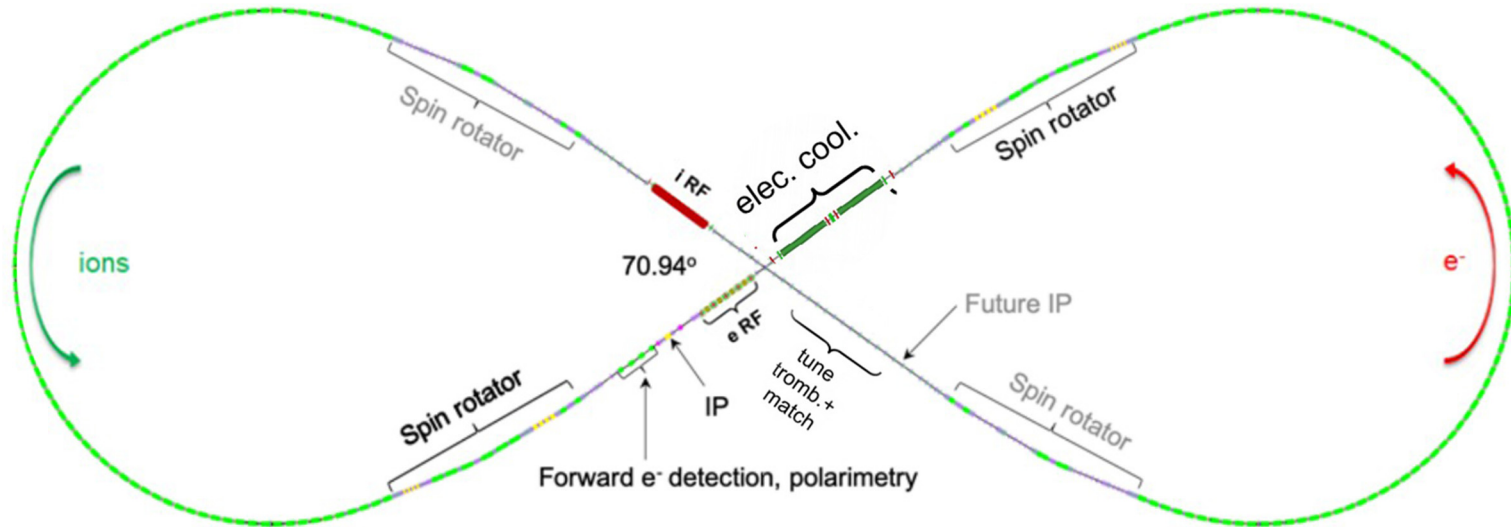
All RT Figure-8 Design

16 GeV/u Lead cell & ring optics

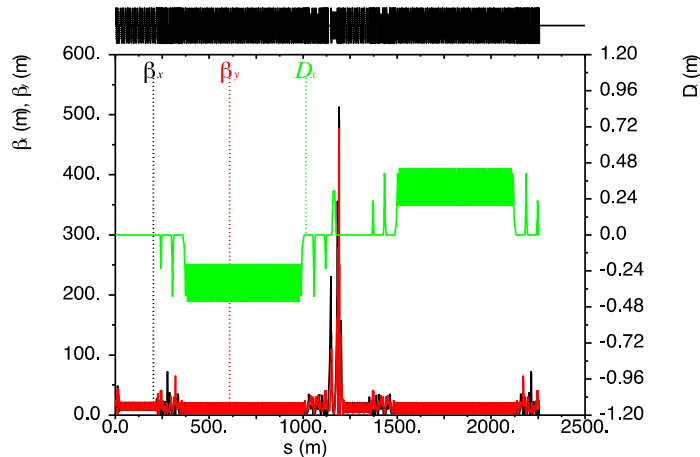
Parameter	Value
Circumference, m	2250.4
Maximum β_x , m	85
Maximum β_y , m	73
Maximum dispersion, m	1.15
γ	18.17
Transition γ	18.6
Quad. half aperture, cm	5
Quad. max. grad. T/m	20
Quad. length in arc, m	1.35
Dipole max. field, T	1.6
Dipole Length, m	5.4
Cell Length, m	17.1



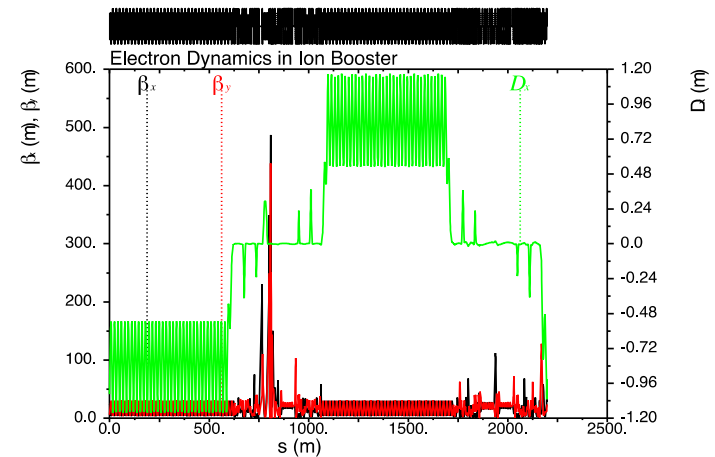
Large Ion Booster as electron Storage Ring



Large Booster with e- and ion insertions



12 GeV e- optics in e-Ring Baseline



12 GeV e- optics in Ion Booster

Conclusions for the 100-GeV Design

- ❑ The Pre-Booster can be small RT race-track ~ 6 GeV - 250 m circumference, solves heavy-ions SC effects, but will require Siberian snake for proton and ^3He spin
- ❑ The Large Booster can be up to 16 GeV/u for all ions in the same footprint as the E-Ring with all RT magnets, the high energy is beneficial for 100-GeV CM and upgrade to 140-GeV
- ❑ The new Large Booster design can be used as e-Ring. It seems feasible from space and beam optics point of view. Practical and operational considerations need to be included ...
- ❑ As the two design approaches are converging, this effort is now coming to a close (thesis work of J. Martinez)

Thank you!